

■ 2.5.5 Handling Error Conditions

Most *Mathematica* functions produce warning messages if you try to use them in inappropriate ways. In writing programs, it is often important to check automatically whether any messages were generated during a particular calculation. If messages were generated, say as a consequence of producing indeterminate numerical results, then the result of the calculation may be meaningless.

`Check[expr, failexpr]` if no messages are generated during the evaluation of `expr`, then return `expr`, otherwise return `failexpr`

`Check[expr, failexpr, s1::t1, s2::t2, ...]`
check only for the messages `si`

Checking for warning messages.

Typing in `0/0` produces a warning message.

```
In[1]:= 0/0

Power::infy: Infinite expression  $\frac{1}{0}$  encountered.

Infinity::indt:
Indeterminate expression 0 ComplexInfinity
encountered.

Out[1]= Indeterminate
```

This defines a function which computes `0/x`. If a message is generated, the function returns the symbol `meaningless`.

```
In[2]:= f[x_] := Check[0/x, meaningless]
```

No message is generated, so the function returns `0/1`.

```
In[3]:= f[1]

Out[3]= 0
```

This produces an error message, and so returns the second argument of `Check`.

```
In[4]:= f[0]

Power::infy: Infinite expression  $\frac{1}{0}$  encountered.

Infinity::indt:
Indeterminate expression 0 ComplexInfinity
encountered.

Out[4]= meaningless
```

`Check` usually tests for all messages, including ones whose output has been suppressed using `Off`.

In some cases, however, you may want to test only for a specific set of messages,

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

say ones associated with numerical overflow. You can do this by explicitly telling **Check** the names of the messages you want to look for.